

Working with Text Files Commands

Nano

```
chunduru@chunduru-virtual-machine:~$ nano text.txt
chunduru@chunduru-virtual-machine:~$ cat text.txt
Karthik
sai
hari

chunduru@chunduru-virtual-machine:~$
```

- Description: A user-friendly text editor.
- Example: `nano filename.txt` - Opens `filename.txt` in nano editor.

Vimtutor

```
=====
=  Welcome to the VIM Tutor - Version 1.7  =
=====

Vim is a very powerful editor that has many commands, too many to
explain in a tutor such as this. This tutor is designed to describe
enough of the commands that you will be able to easily use Vim as
an all-purpose editor.

The approximate time required to complete the tutor is 30 minutes,
depending upon how much time is spent with experimentation.

ATTENTION:
The commands in the lessons will modify the text. Make a copy of this
file to practice on (if you started "vimtutor" this is already a copy).

It is important to remember that this tutor is set up to teach by
use. That means that you need to execute the commands to learn them
properly. If you only read the text, you will forget the commands!

Now, make sure that your Caps-Lock key is NOT depressed and press
the  j  key enough times to move the cursor so that lesson 1.1
completely fills the screen.

=====
Lesson 1.1: MOVING THE CURSOR

** To move the cursor, press the h,j,k,l keys as indicated. **
      ^
      k
    < h       l >      Hint: The h key is at the left and moves left.
      j           The l key is at the right and moves right.
      v           The j key looks like a down arrow.

1. Move the cursor around the screen until you are comfortable.

2. Hold down the down key (j) until it repeats.
   Now you know how to move to the next lesson.

3. Using the down key, move to lesson 1.2.

NOTE: If you are ever unsure about something you typed, press <ESC> to place
"/tmp/tutorSRpLIE" 972 lines, 33583 bytes
```

- Description: A tutorial program for learning VIM, an advanced text editor.
- Example: `vimtutor` - Starts the VIM tutorial.

Vi

```
chunduru@chunduru-virtual-machine:~$ vi text.txt
chunduru@chunduru-virtual-machine:~$ cat text.txt
Karthik
sai
hari
```

- Description: The original text editor, with basic features.
- Example: `vi filename.txt` - Opens `filename.txt` in VI editor.

Vim

```
chunduru@chunduru-virtual-machine:~$ vim text.txt
chunduru@chunduru-virtual-machine:~$ cat text.txt
Karthik
sai
hari
```

- Description: An improved version of VI with more features.
- Example: `vim filename.txt` - Opens `filename.txt` in VIM editor.

More

```
chunduru@chunduru-virtual-machine:~$ more text.txt
Karthik
sai
hari

chunduru@chunduru-virtual-machine:~$ more /var/log/syslog
Aug 9 18:15:22 chunduru-virtual-machine systemd[1]: rsyslog.service: Sent signal SIGHUP to main process 786 (rsyslogd) on client request.
Aug 9 18:15:22 chunduru-virtual-machine NetworkManager[745]: <Info> [1786279522.2641] Loaded device plugin: NMBlueZManager (/usr/lib/x86_64-linux-gnu/NetworkManager/1.36.6/libnm-device-plugin-bluetooth.so)
Aug 9 18:15:22 chunduru-virtual-machine dbus-daemon[742]: [system] Activation via systemd failed for unit 'dbus-org.freedesktop.network1.service': Unit dbus-org.freedesktop.network1.service not found.
Aug 9 18:15:22 chunduru-virtual-machine networkd-dispatcher[1942]: WARNING: systemd-networkd is not running, output will be incomplete.
Aug 9 18:15:22 chunduru-virtual-machine systemd[1]: Started GNOME Display Manager.
Aug 9 18:15:22 chunduru-virtual-machine networkd-dispatcher[764]: ERROR:Unknown state for Interface NetworkctlListState(idx=1, name='lo', type='loopback', operational='n/a', administrative='unmanaged')
n/a
Aug 9 18:15:22 chunduru-virtual-machine networkd-dispatcher[764]: Traceback (most recent call last):
Aug 9 18:15:22 chunduru-virtual-machine networkd-dispatcher[764]:   File "/usr/bin/networkd-dispatcher", line 298, in trigger_all
Aug 9 18:15:22 chunduru-virtual-machine networkd-dispatcher[764]:     self.handle_state(iface_name,
Aug 9 18:15:22 chunduru-virtual-machine networkd-dispatcher[764]:   File "/usr/bin/networkd-dispatcher", line 348, in handle_state
Aug 9 18:15:22 chunduru-virtual-machine networkd-dispatcher[764]:     raise UnknownState(operational_state)
Aug 9 18:15:22 chunduru-virtual-machine networkd-dispatcher[764]: UnknownState: n/a
Aug 9 18:15:22 chunduru-virtual-machine networkd-dispatcher[764]: ERROR:Unknown state for Interface NetworkctlListState(idx=2, name='ens33', type='ether', operational='n/a', administrative='unmanaged')
n/a
Aug 9 18:15:22 chunduru-virtual-machine networkd-dispatcher[764]: Traceback (most recent call last):
Aug 9 18:15:22 chunduru-virtual-machine networkd-dispatcher[764]:   File "/usr/bin/networkd-dispatcher", line 298, in trigger_all
Aug 9 18:15:22 chunduru-virtual-machine networkd-dispatcher[764]:     self.handle_state(iface_name,
Aug 9 18:15:22 chunduru-virtual-machine networkd-dispatcher[764]:   File "/usr/bin/networkd-dispatcher", line 348, in handle_state
Aug 9 18:15:22 chunduru-virtual-machine networkd-dispatcher[764]:     raise UnknownState(operational_state)
Aug 9 18:15:22 chunduru-virtual-machine networkd-dispatcher[764]: UnknownState: n/a
Aug 9 18:15:22 chunduru-virtual-machine NetworkManager[745]: <Info> [1786279522.2529] Loaded device plugin: NMTeamFactory (/usr/lib/x86_64-linux-gnu/NetworkManager/1.36.6/libnm-device-plugin-team.so)
Aug 9 18:15:22 chunduru-virtual-machine systemd[1]: Started Dispatcher daemon for systemd-networkd.
Aug 9 18:15:22 chunduru-virtual-machine systemd[1]: logrotate.service: Deactivated successfully.
Aug 9 18:15:22 chunduru-virtual-machine systemd[1]: Finished Rotate log files.
Aug 9 18:15:22 chunduru-virtual-machine NetworkManager[745]: <Info> [1786279522.2664] Loaded device plugin: NMAtmManager (/usr/lib/x86_64-linux-gnu/NetworkManager/1.36.6/libnm-device-plugin-atsl.so)
Aug 9 18:15:22 chunduru-virtual-machine NetworkManager[745]: <Info> [1786279522.2669] manager: rtkill: WL-Fi enabled by radio killswitch; enabled by state file
Aug 9 18:15:22 chunduru-virtual-machine NetworkManager[745]: <Info> [1786279522.2670] manager: rtkill: WWAN enabled by radio killswitch; enabled by state file
Aug 9 18:15:22 chunduru-virtual-machine NetworkManager[745]: <Info> [1786279522.2670] manager: Networking is enabled by state file
Aug 9 18:15:22 chunduru-virtual-machine atsd[1853]: also-lib main.c:1412:(snd_use_case_mgr_open) error: failed to import hwd use case configuration -2
Aug 9 18:15:22 chunduru-virtual-machine dbus-daemon[742]: [system] Activating via systemd: service name='org.freedesktop.nm_dispatcher' unit='dbus-org.freedesktop.nm_dispatcher.service' requested by '':
```

- Description: A pager to view the contents of text files, one screen at a time.
- Example: `more filename.txt` - Views the content of `filename.txt`.

Less

```
chunduru@chunduru-virtual-machine:~$ less text.txt
chunduru@chunduru-virtual-machine:~$ cat text.txt
Karthik
sai
hari
```

- Description: Similar to more, but provides more control over viewing text files.
- Example: `less filename.txt` - Opens `filename.txt` for viewing with navigation options.

Head

```
chunduru@chunduru-virtual-machine:~$ head text.txt
Karthik
sai
hari
```

- Description: Displays the first few lines of a text file.
- Example: `head filename.txt` - Shows the top lines of `filename.txt`.

Tail

```
chunduru@chunduru-virtual-machine:~$ tail text.txt

ravi
hari
vamsi
```

- Description: Displays the last few lines of a text file.
- Example: `tail filename.txt` - Shows the last lines of `filename.txt`.

Cat

```
chunduru@chunduru-virtual-machine:~$ cat text.txt
Karthik
sai
ganesh

ravi
hari
vamsi
```

- Description: Concatenates and displays the entire content of text files.
- Example: `cat filename.txt` - Displays the full content of `filename.txt`.

cat -A

```
chunduru@chunduru-virtual-machine:~$ cat -A text.txt
Karthik$
sai$
ganesh$
karthik$
ravi$
```

- Description: Similar to cat, but also displays special characters.
- Example: `cat -A filename.txt` - Shows the content of `filename.txt`, including special characters.

Tac

```
chunduru@chunduru-virtual-machine:~$ cat text.txt
line 1
line 2
line 3
line 4
chunduru@chunduru-virtual-machine:~$ tac text.txt
line 4
line 3
line 2
line 1
```

- Description: Prints the contents of a text file in reverse order.
- Example: `tac filename.txt` - Displays `filename.txt` content backwards.

Grep

```
chunduru@chunduru-virtual-machine:~$ grep "Linux" text.txt
Hello Linux
Linux is operating system
chunduru@chunduru-virtual-machine:~$
```

- Description: Searches for text in files or command output.
- Example: `grep 'pattern' filename.txt` - Searches for 'pattern' in `filename.txt`.

Advanced Text File Processing Commands

grep:

```
chunduru@chunduru-virtual-machine:~$ grep "Linux" text.txt
Hello Linux
Linux is operating system
chunduru@chunduru-virtual-machine:~$
```

- Command: "grep"
- Example: "grep 'pattern' file.txt"
- Description: Filters text in files by searching for specific patterns.

awk:

```
chunduru@chunduru-virtual-machine:~$ awk '{print $1}' text.txt
karthik
ravi
hari
chunduru@chunduru-virtual-machine:~$
```

- Command: "awk"
- Example: "awk '/pattern/ {print}' file.txt"
- Description: Filters and processes text, with additional features for text manipulation.

sed:

```
chunduru@chunduru-virtual-machine:~$ sed 's/Linux/Devops/g' text.txt
Hello Devops
Devops is easy
I love Devops
```

- Command: "sed"
- Example: "sed 's/old/new/' file.txt"
- Description: A stream editor used for modifying text files directly from the command line.

tr:

```
chunduru@chunduru-virtual-machine:~$ echo "HELLO LINX" | tr 'a-z' 'A-Z'
HELLO LINX
chunduru@chunduru-virtual-machine:~$ echo "HELLO LINX" | tr 'A-Z' 'a-z'
hello linx
chunduru@chunduru-virtual-machine:~$
```

- Command: "tr"
- Example: "tr '[:lower:]' '[:upper:]' < file.txt"
- Description: Translates or deletes characters from input text.

cut:

```
chunduru@chunduru-virtual-machine:~$ echo "Devops" | cut -c 1-3
Dev
```

- Command: "cut"
- Example: "cut -d':' -f1 file.txt"
- Description: Extracts sections from each line of files, such as columns from a table.

sort:

```
chunduru@chunduru-virtual-machine:~$ cat text.txt
ravi
ganesh
anil
suresh

chunduru@chunduru-virtual-machine:~$ sort text.txt
anil
ganesh
ravi
suresh
chunduru@chunduru-virtual-machine:~$
```

- Command: "sort"
- Example: "sort file.txt"
- Description: Sorts text files in alphabetical or numerical order, as specified.

Connecting to a Server Commands

su Command

```
chunduru@chunduru-virtual-machine:~$ su techie
Password:
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

techie@chunduru-virtual-machine:/home/chunduru$ whoami
techie
techie@chunduru-virtual-machine:/home/chunduru$
```

- **Example:** `su username`
- **Description:** This command is used to switch user identity in a Unix-like operating system. When you type `su` followed by a username, it prompts you for the password of that user and, if entered correctly, logs you in as that user.

sudo Command

```
techie@chunduru-virtual-machine:/$ sudo apt update
[sudo] password for techie:
Sorry, try again.
[sudo] password for techie:
Hit:1 http://in.archive.ubuntu.com/ubuntu jammy InRelease
Hit:2 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:3 http://in.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:4 http://in.archive.ubuntu.com/ubuntu jammy-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
techie@chunduru-virtual-machine:/$
```

- **Example:** `sudo apt update`
- **Description:** The `sudo` command allows a permitted user to execute a command as the superuser or another user, as specified in the `sudoers` file. This example updates the list of available packages and their versions, but it requires administrative privileges, hence the use of `sudo`.

ssh Command

- **Example:** `ssh username@hostname`
- **Description:** `ssh` stands for Secure Shell. It's used to connect and log into a remote machine securely. The example shows how to connect to a remote host where `username` is your username on the remote machine, and `hostname` is the address of the remote machine.

scp Command

- **Example:** `scp file.txt username@hostname:/path/to/destination`

- Description: The `scp` command is used for secure copy from one machine to another. In the example, `file.txt` is being copied to a remote host. The `username@hostname` specifies the user and host of the remote machine, and `/path/to/destination` is where the file will be placed on the remote machine.

Working with the Bash Shell commands

Tee

```
chunduru@chunduru-virtual-machine:~$ ls | tee text.txt
1
Desktop
Documents
Downloads
extract
karthik
linl.txt
Music
Pictures
Public
snap
techie.tar
Templates
test
test-tar
test.tar
test.tar.gz
text.txt
Videos
```

- Example: `ls | tee file.txt`
- Description: Used in pipes to direct the output of a command both to a file and the next command in the pipe.

History

```
chunduru@chunduru-virtual-machine:~$ history
```

```
1 /
2 cd /
3 echo $
4 echo $0
5 cd /
6 pwd
7 clear
8 pwd
9 ls
10 ll
11 cd
12 clear
13 cd ~
14 pwd
15 ls
16 cd desktop
17 pwd
18 ls
19 cd desktop
20 cd ~
21 pwd
22 ls
23 cd Desktop/
24 ls
25 pwd
26 cd..
27 ls
28 pwd
29 cd ..
30 pwd
31 cd .. / ..
32 cd Desktop/
33 pwd
34 cd ../..
35 pwd
36 cd chunduru/Desktop/
37 pwd
38 cd ~
39 pwd
```

- Example: `history`
- Description: Displays a list of previously executed commands.

`history -d`

```
chunduru@chunduru-virtual-machine:~$ history -d 427
```

```
chunduru@chunduru-virtual-machine:~$ history
```

```
1 /
2 cd /
3 echo $
4 echo $0
5 cd /
6 pwd
7 clear
8 pwd
9 ls
10 ll
11 cd
12 clear
13 cd ~
14 pwd
15 ls
16 cd desktop
17 pwd
18 ls
19 cd desktop
20 cd ~
21 pwd
22 ls
23 cd Desktop/
24 ls
25 pwd
26 cd..
27 ls
```

- Example: `history -d 1023`
- Description: Deletes a specific command from the history, where `1023` is the history entry number you want to remove.

Alias

```
chunduru@chunduru-virtual-machine:~$ alias ll='ls -l'
chunduru@chunduru-virtual-machine:~$ ll
total 96
-rw-rw-r-- 1 chunduru chunduru 11 Aug 8 11:58 1
drwxr-xr-x 2 chunduru chunduru 4096 Aug 8 12:14 Desktop
drwxr-xr-x 2 chunduru chunduru 4096 Aug 4 11:20 Documents
drwxr-xr-x 2 chunduru chunduru 4096 Aug 4 11:20 Downloads
drwxrwxr-x 3 chunduru chunduru 4096 Aug 8 17:22 extract
drwxrwxr-x 2 chunduru chunduru 4096 Aug 8 12:08 karthik
lrwxrwxrwx 1 chunduru chunduru 9 Aug 8 12:53 linl.text -> file.text
drwxr-xr-x 2 chunduru chunduru 4096 Aug 4 11:20 Music
drwxr-xr-x 2 chunduru chunduru 4096 Aug 4 11:20 Pictures
drwxr-xr-x 2 chunduru chunduru 4096 Aug 4 11:20 Public
drwx----- 4 chunduru chunduru 4096 Aug 8 13:34 snap
-rw-rw-r-- 1 chunduru chunduru 20480 Aug 8 14:16 techie.tar
drwxr-xr-x 2 chunduru chunduru 4096 Aug 4 11:20 Templates
drwxrwxr-x 2 chunduru chunduru 4096 Aug 8 17:19 test
drwxrwxr-x 2 chunduru chunduru 4096 Aug 8 14:13 test-tar
-rw-rw-r-- 1 chunduru chunduru 10240 Aug 8 17:19 test.tar
-rw-rw-r-- 1 chunduru chunduru 153 Aug 8 17:23 test.tar.gz
-rw-rw-r-- 1 chunduru chunduru 155 Aug 9 19:26 text.txt
drwxr-xr-x 2 chunduru chunduru 4096 Aug 4 11:20 Videos
chunduru@chunduru-virtual-machine:~$
```

- Example: `alias ll='ls -l'`
- Description: Creates a shortcut for a longer command. In this example, typing `ll` will execute `ls -l`.

Export

```
chunduru@chunduru-virtual-machine:~$ export NAME=karthik
chunduru@chunduru-virtual-machine:~$ echo $NAME
karthik
chunduru@chunduru-virtual-machine:~$
```

- Example: `export VAR=value`
- Description: Sets an environment variable `VAR` to a value `value` and makes it available to subshells.

User and Group Management and Permissions Commands

useradd: Adds users.

```
chunduru@chunduru-virtual-machine:~$ sudo useradd techie
useradd: user 'techie' already exists
chunduru@chunduru-virtual-machine:~$ su - techie
Password:
techie@chunduru-virtual-machine:~$ whoami
techie
techie@chunduru-virtual-machine:~$
```

- Example: `useradd johndoe`

- Description: Creates a new user account named 'johndoe'.

usermod: Modifies user properties.

```
chundurur@chundurur-virtual-machine:~$ sudo usermod -aG sudo Developer
chundurur@chundurur-virtual-machine:~$ groups Developer
Developer : techie sudo
chundurur@chundurur-virtual-machine:~$ su -Developer
su: invalid option -- 'D'
Try 'su --help' for more information.
chundurur@chundurur-virtual-machine:~$ su - Developer
Password:
Developer@chundurur-virtual-machine:~$ whoami
Developer
Developer@chundurur-virtual-machine:~$ su - developer
su: user developer does not exist or the user entry does not contain all the required fields
Developer@chundurur-virtual-machine:~$ su - Developer
Password:
Developer@chundurur-virtual-machine:~$ whoami
Developer
```

- Example: `usermod -L johndoe`
- Description: Locks the user account 'johndoe'.

userdel: Deletes users.

```
Developer@chundurur-virtual-machine:~$ userdel Developer
userdel: user Developer is currently used by process 4925
Developer@chundurur-virtual-machine:~$ exit
logout
Developer@chundurur-virtual-machine:~$ sudo userdel Developer
[sudo] password for Developer:
userdel: user Developer is currently used by process 4925
Developer@chundurur-virtual-machine:~$ exit
logout
chundurur@chundurur-virtual-machine:~$ whoami
chundurur
chundurur@chundurur-virtual-machine:~$ userdel Developer
userdel: Permission denied.
userdel: cannot lock /etc/passwd; try again later.
chundurur@chundurur-virtual-machine:~$ sudo userdel Developer
chundurur@chundurur-virtual-machine:~$ id Developer
id: 'Developer': no such user
chundurur@chundurur-virtual-machine:~$
```

- Example: `userdel johndoe`
- Description: Removes the user account 'johndoe'.

groupadd: Adds groups.

```
chundurur@chundurur-virtual-machine:~$ sudo groupadd developers
groupadd: group 'developers' already exists
chundurur@chundurur-virtual-machine:~$ sudo usermod -aG developers chundurur
chundurur@chundurur-virtual-machine:~$ groups chundurur
chundurur : chundurur adm cdrom sudo dip plugdev lpadmin lxd sambashare developers
chundurur@chundurur-virtual-machine:~$
```

- Example: `groupadd developers`
- Description: Creates a new group named 'developers'.

groupmod: Modifies group properties.

```
chundurur@chundurur-virtual-machine:~$ sudo groupmod -n developers_team developers
chundurur@chundurur-virtual-machine:~$ getnet group developers_team
Command 'getnet' not found, did you mean:
  command 'getent' from deb libc-bin (2.35-0ubuntu3.13)
Try: sudo apt install <deb name>
chundurur@chundurur-virtual-machine:~$ getent group developers_team
developers_team:x:1002:chundurur
chundurur@chundurur-virtual-machine:~$ groups chundurur
chundurur : chundurur adm cdrom sudo dip plugdev lpadmin lxd sambashare developers_team
chundurur@chundurur-virtual-machine:~$
```

- Example: `groupmod -n newname oldname`

- Description: Renames a group from 'oldname' to 'newname'.

groupdel: Deletes groups.

```
chunduru@chunduru-virtual-machine:~$ sudo groupdel developers_team
chunduru@chunduru-virtual-machine:~$ getent group developer_team
chunduru@chunduru-virtual-machine:~$
```

- Example: `groupdel developers`
- Description: Removes the group named 'developers'.

id: Shows information about user accounts.

```
chunduru@chunduru-virtual-machine:~$ id chunduru
uid=1000(chunduru) gid=1000(chunduru) groups=1000(chunduru),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),122(lpadmin),135(lxd),136(sambashare)
chunduru@chunduru-virtual-machine:~$
```

- Example: `id johndoe`
- Description: Displays user and group IDs for 'johndoe'.

getent: Gets information from administrative databases.

```
chunduru@chunduru-virtual-machine:~$ getent passwd chunduru
chunduru:x:1000:1000:chunduru venkata karthik,,,:/home/chunduru:/bin/bash
chunduru@chunduru-virtual-machine:~$
```

- Example: `getent passwd johndoe`
- Description: Retrieves user 'johndoe' details from the passwd database.

passwd: Changes passwords.

```
chunduru@chunduru-virtual-machine:~$ passwd chunduru
Changing password for chunduru.
Current password:
New password:
Retype new password:
passwd: password updated successfully
chunduru@chunduru-virtual-machine:~$
```

- Example: `passwd johndoe`
- Description: Sets or changes the password for 'johndoe'.

chage: Modifies password properties.

```
chunduru@chunduru-virtual-machine:~$ sudo chage -l chunduru
Last password change           : Aug 09, 2026
Password expires                : never
Password inactive              : never
Account expires                : never
Minimum number of days between password change : 0
Maximum number of days between password change : 99999
Number of days of warning before password expires : 7
chunduru@chunduru-virtual-machine:~$
```

- Example: `chage -E 2023-12-31 johndoe`
- Description: Sets the expiry date of 'johndoe's password.

w: Shows logged-in users and details.

```
chundurur@chundurur-virtual-machine:~$ w
20:36:54 up 2:21, 1 user, load average: 0.02, 0.04, 0.00
USER      TTY      FROM            LOGIN@   IDLE   JCPU   PCPU   WHAT
chundurur tty2      tty2            18:15    2:21m  0.20s  0.17s  /usr/libexec/gnome-session-binary --session=ubuntu
chundurur@chundurur-virtual-machine:~$
```

- Example: `w`
- Description: Lists users currently logged in with session details.

who: Shows currently logged-in users.

```
chundurur@chundurur-virtual-machine:~$ who
chundurur tty2          2026-08-09 18:15 (tty2)
chundurur@chundurur-virtual-machine:~$
```

- Example: `who`
- Description: Lists users currently logged in with minimal details.

vipw: Edits the passwd file.

```
chundurur@chundurur-virtual-machine:~$ sudo vipw

Select an editor. To change later, run 'select-editor'.
 1. /bin/nano        <---- easiest
 2. /usr/bin/vim.basic
 3. /usr/bin/vim.tiny
 4. /bin/ed

Choose 1-4 [1]:
```

- Example: `vipw`
- Description: Opens the password file for editing.

vigr: Edits the `/etc/group` file.

```
GNU nano 6.2 /etc/group.edi
daemon:x:1:
bin:x:2:
sys:x:3:
adm:x:4:syslog,chundurur
tty:x:5:
disk:x:6:
lp:x:7:
mail:x:8:
news:x:9:
uucp:x:10:
man:x:12:
proxy:x:13:
kmem:x:15:
dialout:x:20:
fax:x:21:
voice:x:22:
cdrom:x:24:chundurur
floppy:x:25:
tape:x:26:
sudo:x:27:chundurur
audio:x:29:pulse
dtp:x:30:chundurur
www-data:x:33:
backup:x:34:
operator:x:37:
list:x:38:
irc:x:39:
src:x:40:
gnats:x:41:
shadow:x:42:
uucp:x:43:
video:x:44:
sasl:x:45:
plugdev:x:46:chundurur
staff:x:50:
games:x:60:
users:x:100:
```

- Example: `vigr`
- Description: Opens the group file for editing.

loginctl: Manages user sessions.

```
chundurur@chundurur-virtual-machine:~$ loginctl
SESSION  UID USER      SEAT  TTY
      2 1000 chundurur seat0 tty2

1 sessions listed.
```

- Example: `loginctl list-sessions`
- Description: Lists current user sessions.

Permissions Management commands

Chown

```
chunduru@chunduru-virtual-machine:~$ sudo chown chunduru techie
chunduru@chunduru-virtual-machine:~$ ls -l techie
-rw-rw-r-- 1 chunduru chunduru 0 Aug  9 20:48 techie
chunduru@chunduru-virtual-machine:~$
```

- Example: `chown user:group filename`
- Description: Changes the ownership of a file or directory. The command alters both user and group ownership. It's a versatile command that eliminates the need for `chgrp` in most cases.

Chgrp

```
chunduru@chunduru-virtual-machine:~$ sudo chgrp sudo techie
chunduru@chunduru-virtual-machine:~$ ls -l techie
-rw-rw-r-- 1 chunduru sudo 0 Aug  9 20:48 techie
chunduru@chunduru-virtual-machine:~$
```

- Example: `chgrp group filename`
- Description: Changes the group ownership of a file or directory. While this command is dedicated to changing group ownership, `chown` can also perform this function.

Chmod

```
chunduru@chunduru-virtual-machine:~$ chmod +x techie
chunduru@chunduru-virtual-machine:~$ ls -l techie
-rwxrwxr-x 1 chunduru sudo 0 Aug  9 20:48 techie
chunduru@chunduru-virtual-machine:~$
```

- Example: `chmod 755 filename`
- Description: Changes the permission mode of a file or directory. It's an essential command for managing file and directory permissions in terms of read, write, and execute rights.

Diff

```
chunduru@chunduru-virtual-machine:~$ diff file1.txt file2.txt
1c1
< hello
---
> hi
chunduru@chunduru-virtual-machine:~$
```

- Example: `diff file1 file2`
- Description: Compares two files and displays the differences between them. This command is

particularly useful in scenarios like checking changes in configuration files or script versions.

find (in the context of set user ID)

```
chunduru@chunduru-virtual-machine:~$ find . -name "*.txt"
./text.txt
./snap/firefox/common/.mozilla/firefox/eaizst2x.default/pkcs11.txt
./file1.txt
./file2.txt
./.cache/tracker3/files/first-index.txt
./.cache/tracker3/files/last-crawl.txt
./.cache/tracker3/files/locale-for-miner-apps.txt
chunduru@chunduru-virtual-machine:~$
```

- Example: `find / -perm /4000`
- Description: Searches for files with set user ID permissions. This is important for security and system administration, to locate files that might have special privileges.

Umask

```
chunduru@chunduru-virtual-machine:~$ umask
0002
chunduru@chunduru-virtual-machine:~$
```

- Example: `umask 022`
- Description: Sets the default permissions for newly created files and directories. It's a foundational command for establishing baseline security and access controls.

Storage Management Essentials commands

Lsblk

```
chunduru@chunduru-virtual-machine:~$ lsblk
NAME        MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
fd0          2:0    1     4K  0 disk
loop0        7:0     0     4K  1 loop /snap/bare/5
loop1        7:1     0  74.3M  1 loop /snap/core22/1612
loop2        7:2     0 271.2M  1 loop /snap/firefox/4848
loop3        7:3     0 505.1M  1 loop /snap/gnome-42-2204/176
loop4        7:4     0 531.5M  1 loop /snap/gnome-42-2204/263
loop5        7:5     0  12.9M  1 loop /snap/snap-store/1113
loop6        7:6     0  91.7M  1 loop /snap/gtk-common-themes/1535
loop7        7:7     0  12.2M  1 loop /snap/snap-store/1216
loop8        7:8     0  38.8M  1 loop /snap/snapd/21759
loop9        7:9     0  50.1M  1 loop /snap/snapd/27591
loop10       7:10    0   500K  1 loop /snap/snapd-desktop-integration/178
loop11       7:11    0   66.8M  1 loop /snap/core24/1643
loop12       7:12    0   828K  1 loop /snap/snapd-desktop-integration/391
loop13       7:13    0 614.5M  1 loop /snap/gnome-46-2404/164
loop14       7:14    0  402M  1 loop /snap/mesa-2404/1839
sda          8:0     0    20G  0 disk
├─sda1       8:1     0     1M  0 part
├─sda2       8:2     0   513M  0 part /boot/efi
└─sda3       8:3     0  19.5G  0 part /
sr0         11:0     1 154.7M  0 rom  /media/chunduru/CDROM
sr1         11:1     1   4.4G  0 rom  /media/chunduru/Ubuntu 22.04.5 LTS amd64
chunduru@chunduru-virtual-machine:~$
```

- Description: Lists all block devices.
- Example: `lsblk`
- This command displays information about all block devices connected to the system, such as hard drives and USB drives.

Fdisk

```
chunduru@chunduru-virtual-machine:~$ sudo fdisk -l
Disk /dev/loop0: 4 KiB, 4096 bytes, 8 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/loop1: 74.27 MiB, 77881344 bytes, 152112 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/loop2: 271.24 MiB, 284413952 bytes, 555496 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/loop3: 505.09 MiB, 529625088 bytes, 1034424 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/loop4: 531.49 MiB, 557309952 bytes, 1088496 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/loop5: 12.93 MiB, 13553664 bytes, 26472 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
```

- Description: Manages partitions on an MBR disk.
- Example: `fdisk /dev/sda`
- Used for creating and managing disk partitions on an MBR (Master Boot Record) formatted disk.

Gdisk

```
chundurur@chundurur-virtual-machine:~$ sudo gdisk
GPT fdisk (gdisk) version 1.0.8

Type device filename, or press <Enter> to exit:
```

- Description: Manages partitions on a GPT disk.
- Example: `gdisk /dev/sda`
- Similar to `fdisk` but for GPT (GUID Partition Table) formatted disks, which are more modern.

Mkfs

```
chundurur@chundurur-virtual-machine:~$ sudo mkfs -t ext4 /dev/sdb1
mke2fs 1.46.5 (30-Dec-2021)
The file /dev/sdb1 does not exist and no size was specified.
chundurur@chundurur-virtual-machine:~$
```

- Description: Creates a filesystem.
- Example: `mkfs.ext4 /dev/sda1`
- A generic utility to format a partition with a specific file system, such as ext4, xfs, or btrfs.

Mount

```
chundurur@chundurur-virtual-machine:~$ sudo mkdir /mnt/nydisk
chundurur@chundurur-virtual-machine:~$ lsblk -f
```

NAME	FSTYPE	FSVER	LABEL	UUID	FSAVAIL	FSUSEX	MOUNTPOINTS
fd0							
loop0	squashfs	4.0			0	100%	/snap/bare/5
loop1	squashfs	4.0			0	100%	/snap/core22/1612
loop2	squashfs	4.0			0	100%	/snap/firefox/4848
loop3	squashfs	4.0			0	100%	/snap/gnome-42-2204/176
loop4	squashfs	4.0			0	100%	/snap/gnome-42-2204/263
loop5	squashfs	4.0			0	100%	/snap/snap-store/1113
loop6	squashfs	4.0			0	100%	/snap/gtk-common-themes/1535
loop7	squashfs	4.0			0	100%	/snap/snap-store/1216
loop8	squashfs	4.0			0	100%	/snap/snapd/21759
loop9	squashfs	4.0			0	100%	/snap/snapd/27591
loop10	squashfs	4.0			0	100%	/snap/snapd-desktop-integration/178
loop11	squashfs	4.0			0	100%	/snap/core24/1643
loop12	squashfs	4.0			0	100%	/snap/snapd-desktop-integration/391
loop13	squashfs	4.0			0	100%	/snap/gnome-46-2404/164
loop14	squashfs	4.0			0	100%	/snap/mesa-2404/1839
sda							
-sda1							
-sda2	vfat	FAT32		89D9-61AD	505.9M	1%	/boot/efi
-sda3	ext4	1.0		b22b464c-b5bd-41e0-bcca-259bfd437dc	4.3G	72%	/
sr0	iso9660		CDROM	2026-08-04-10-54-42-00	0	100%	/media/chundurur/CDROM
sr1	iso9660		Joliet Extension Ubuntu 22.04.5 LTS amd64	2024-09-11-14-37-52-00	0	100%	/media/chundurur/Ubuntu 22.04.5 LTS amd64

```
chundurur@chundurur-virtual-machine:~$ sudo mount /dev/sdb1 /mnt/nydisk
mount: /mnt/nydisk: mount point does not exist.
chundurur@chundurur-virtual-machine:~$ findmnt
```

TARGET	SOURCE	FSTYPE	OPTIONS
/	/dev/sda3	ext4	rw,relatime,errors=remount-ro
/sys	sysfs	sysfs	rw,nosuid,nodev,noexec,relatime
/sys/kernel/security	securityfs	securityfs	rw,nosuid,nodev,noexec,relatime
/sys/fs/cgroup	cgroup2	cgroup2	rw,nosuid,nodev,noexec,relatime,nsdelegate,memory_recursiveprot
/sys/fs/pstore	pstore	pstore	rw,nosuid,nodev,noexec,relatime
/sys/fs/bpf	bpf	bpf	rw,nosuid,nodev,noexec,relatime,mode=700
/sys/kernel/debug	debugfs	debugfs	rw,nosuid,nodev,noexec,relatime
/sys/kernel/tracing	tracefs	tracefs	rw,nosuid,nodev,noexec,relatime

- Description: Attaches a filesystem to the file tree.
- Example: `mount /dev/sda1 /mnt`
- This command is used to mount a filesystem found on a device to a specific directory in the file tree.

Findmnt

```
chundurur@chundurur-virtual-machine:~$ findmnt
TARGET                                SOURCE                                FSTYPE                                OPTIONS
/                                     /dev/sda3                            ext4                                  rw,relatime,errors=remount-ro
/sys                                  sysfs                                  sysfs                                  rw,nosuid,nodev,noexec,relatime
/sys/kernel/security                 securityfs                             securityfs                           rw,nosuid,nodev,noexec,relatime
/sys/fs/cgroup                       cgroup2                               cgroup2                             rw,nosuid,nodev,noexec,relatime,nsdelegate,memory_recursiveprot
/sys/fs/pstore                       pstore                               pstore                             rw,nosuid,nodev,noexec,relatime
/sys/fs/bpf                          bpf                                   bpf                                  rw,nosuid,nodev,noexec,relatime,mode=700
/sys/kernel/debug                    debugfs                               debugfs                              rw,nosuid,nodev,noexec,relatime
/sys/kernel/tracing                  tracefs                               tracefs                              rw,nosuid,nodev,noexec,relatime
/sys/fs/fuse/connections              fusectl                               fusectl                              rw,nosuid,nodev,noexec,relatime
/sys/kernel/config                   configfs                              configfs                             rw,nosuid,nodev,noexec,relatime
/proc                                 proc                                  proc                                 rw,nosuid,nodev,noexec,relatime
/proc/sys/fs/binfmt_misc             binfmt_misc                           binfmt_misc                         rw,relatime,fd=29,pgrp=1,timeout=0,minproto=5,maxproto=5,direct,pipe_ino=12390
/proc/sys/fs/binfmt_misc             binfmt_misc                           binfmt_misc                         rw,nosuid,nodev,noexec,relatime
/dev                                 udev                                  udev                                 rw,nosuid,nodev,noexec,relatime,size=939864k,nr_inodes=234966,mode=755,inode64
/dev/pts                             devpts                                devpts                               rw,nosuid,noexec,relatime,gid=5,mode=620,ptmxmode=000
/dev/shm                             tmpfs                                 tmpfs                               rw,nosuid,nodev,inode64
/dev/hugepages                       hugetlbfs                             hugetlbfs                           rw,relatime,pagesize=2M
/dev/nqueue                           nqueue                               nqueue                              rw,nosuid,nodev,noexec,relatime
/run                                  tmpfs                                 tmpfs                               rw,nosuid,nodev,noexec,relatime,size=196632k,mode=755,inode64
/run/lock                             tmpfs                                 tmpfs                               rw,nosuid,nodev,noexec,relatime,size=5120k,inode64
/run/vmblock-fuse                    fuse.vmblock                         fuse.vmblock-fuse                  rw,relatime,user_id=0,group_id=0,default_permissions,allow_other
/run/credentials/systemd-sysusers.service ramfs                                ramfs                               ro,nosuid,nodev,noexec,relatime,mode=700
/run/snappd/ns                       tmpfs[snapp/ns]                     tmpfs                               rw,nosuid,nodev,noexec,relatime,size=196632k,mode=755,inode64
/run/snappd/ns/mesa-2404.mnt          nsfs[mnt:[4026532715]]              nsfs                                rw
/run/snappd/ns/snapp-desktop-integration.mnt nsfs[mnt:[4026532711]]              nsfs                                rw
/run/user/1000                       tmpfs                                tmpfs                               rw,nosuid,nodev,relatime,size=196628k,nr_inodes=49157,mode=700,uid=1000,gid=1000,inode64
/run/user/1000/doc                   fuse.gvfsd-fuse                     fuse.gvfsd-fuse                    rw,nosuid,nodev,relatime,user_id=1000,group_id=1000
/run/user/1000/gvfs                  fuse.gvfsd-fuse                     fuse.gvfsd-fuse                    rw,nosuid,nodev,relatime,user_id=1000,group_id=1000
/snap/bare/5                         ro,nodev,relatime,errors=continue,threads=single
/snap/core22/1612                    ro,nodev,relatime,errors=continue,threads=single
/snap/firefox/4848                   ro,nodev,relatime,errors=continue,threads=single
/snap/gnome-42-2204/176              ro,nodev,relatime,errors=continue,threads=single
```

- Description: Lists mounted filesystems.
- Example: `findmnt`
- Provides a detailed overview of all the filesystems currently mounted in the system.

Umount

```
chundurur@chundurur-virtual-machine:~$ sudo umount /mnt/mydisk
umount: /mnt/mydisk: not mounted.
```

- Description: Detaches the filesystem from the file tree.
- Example: `umount /mnt`
- Used to unmount a previously mounted filesystem.

Lsof

```
chundurur@chundurur-virtual-machine:~$ lsof techie
chundurur@chundurur-virtual-machine:~$ sudo lsof -i :80
COMMAND PID USER FD TYPE DEVICE SIZE/OFF NODE NAME
apache2 975 root 4u IPv6 17570 0t0 TCP *:http (LISTEN)
apache2 977 www-data 4u IPv6 17570 0t0 TCP *:http (LISTEN)
apache2 1008 www-data 4u IPv6 17570 0t0 TCP *:http (LISTEN)
chundurur@chundurur-virtual-machine:~$
```

- Description: Lists open files and the processes using them.
- Example: `lsof /dev/sda1`
- Useful for diagnosing why a device might be busy, particularly if trying to unmount it and receiving a "device is busy" message.

Managing Networking commands

Command: `ip a`

```
chunduru@chunduru-virtual-machine:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:d0:4e:1d brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.85.128/24 brd 192.168.85.255 scope global dynamic noprefixroute ens33
        valid_lft 1176sec preferred_lft 1176sec
    inet6 fe80::4a42:37be:c1f7:cac5/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
chunduru@chunduru-virtual-machine:~$
```

- Example: `ip a`
- Description: Prints current IP address information, showing the IP addresses assigned to all network interfaces.

Command: `ip route show`

```
chunduru@chunduru-virtual-machine:~$ ip route show
default via 192.168.85.2 dev ens33 proto dhcp metric 100
169.254.0.0/16 dev ens33 scope link metric 1000
192.168.85.0/24 dev ens33 proto kernel scope link src 192.168.85.128 metric 100
chunduru@chunduru-virtual-machine:~$
```

- Example: `ip route show`
- Description: Displays the routing table, including the default route, which is essential for understanding network routes.

Command: `ping`

```
chunduru@chunduru-virtual-machine:~$ ping google.com
PING google.com (142.251.222.206) 56(84) bytes of data:
64 bytes from pnmaaa-as-in-f14.1e100.net (142.251.222.206): icmp_seq=1 ttl=128 time=20.7 ms
64 bytes from pnmaaa-as-in-f14.1e100.net (142.251.222.206): icmp_seq=2 ttl=128 time=20.4 ms
64 bytes from pnmaaa-as-in-f14.1e100.net (142.251.222.206): icmp_seq=3 ttl=128 time=17.9 ms
64 bytes from hkg07s55-in-f14.1e100.net (142.251.222.206): icmp_seq=4 ttl=128 time=20.1 ms
64 bytes from pnmaaa-as-in-f14.1e100.net (142.251.222.206): icmp_seq=5 ttl=128 time=19.6 ms
64 bytes from hkg07s55-in-f14.1e100.net (142.251.222.206): icmp_seq=6 ttl=128 time=20.3 ms
64 bytes from hkg07s55-in-f14.1e100.net (142.251.222.206): icmp_seq=7 ttl=128 time=19.9 ms
64 bytes from hkg07s55-in-f14.1e100.net (142.251.222.206): icmp_seq=8 ttl=128 time=21.2 ms
64 bytes from hkg07s55-in-f14.1e100.net (142.251.222.206): icmp_seq=9 ttl=128 time=20.7 ms
64 bytes from hkg07s55-in-f14.1e100.net (142.251.222.206): icmp_seq=10 ttl=128 time=19.6 ms
64 bytes from hkg07s55-in-f14.1e100.net (142.251.222.206): icmp_seq=11 ttl=128 time=19.9 ms
64 bytes from pnmaaa-as-in-f14.1e100.net (142.251.222.206): icmp_seq=12 ttl=128 time=19.2 ms
64 bytes from hkg07s55-in-f14.1e100.net (142.251.222.206): icmp_seq=13 ttl=128 time=19.6 ms
64 bytes from hkg07s55-in-f14.1e100.net (142.251.222.206): icmp_seq=14 ttl=128 time=19.5 ms
64 bytes from pnmaaa-as-in-f14.1e100.net (142.251.222.206): icmp_seq=15 ttl=128 time=20.4 ms
64 bytes from pnmaaa-as-in-f14.1e100.net (142.251.222.206): icmp_seq=16 ttl=128 time=18.9 ms
64 bytes from pnmaaa-as-in-f14.1e100.net (142.251.222.206): icmp_seq=17 ttl=128 time=20.7 ms
```

- Example: `ping google.com`
- Description: Tests connectivity to a network host (e.g., a website like google.com) by sending ICMP packets.

Command: ss

```
chundurur@chundurur-virtual-machine:~$ ss -tuln
```

NetId	State	Recv-Q	Send-Q	Local Address:Port	Peer Address:Port	Process
udp	UNCONN	0	0	127.0.0.53%lo:53	0.0.0.0:*	
udp	UNCONN	0	0	0.0.0.0:5353	0.0.0.0:*	
udp	UNCONN	0	0	0.0.0.0:43247	0.0.0.0:*	
udp	UNCONN	0	0	:::46787	:::*	
udp	UNCONN	0	0	:::5353	:::*	
tcp	LISTEN	0	4096	127.0.0.53%lo:53	0.0.0.0:*	
tcp	LISTEN	0	128	127.0.0.1:631	0.0.0.0:*	
tcp	LISTEN	0	70	127.0.0.1:33060	0.0.0.0:*	
tcp	LISTEN	0	151	127.0.0.1:3306	0.0.0.0:*	
tcp	LISTEN	0	100	*:8080	*:*	
tcp	LISTEN	0	128	:::1:631	:::1:*	
tcp	LISTEN	0	511	:::80	:::*	

```
chundurur@chundurur-virtual-machine:~$
```

- Example: `ss -tuln`
- Description: Shows socket statistics. Used to check open ports and listening services on the system.

Command: dig

```
chundurur@chundurur-virtual-machine:~$ dig example.com
```

```
; <<>> DiG 9.18.39-0ubuntu0.22.04.4-Ubuntu <<>> example.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 20815
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;example.com.                IN      A

;; ANSWER SECTION:
example.com.                5       IN      A      104.20.23.154
example.com.                5       IN      A      172.66.147.243

;; Query time: 5 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Sun Aug 09 21:14:48 IST 2026
;; MSG SIZE rcvd: 72

chundurur@chundurur-virtual-machine:~$
```

- Example: `dig example.com`
- Description: Queries DNS information. It's useful for troubleshooting DNS issues or verifying DNS records.

Command: nmap

```
chundurur@chundurur-virtual-machine:~$ nmap --version
```

```
Nmap version 7.80 ( https://nmap.org )
Platform: x86_64-pc-linux-gnu
Compiled with: liblua-5.3.6 openssl-3.0.2 nmap-libssh2-1.8.2 libz-1.2.11 libpcrc-8.39 libpcap-1.10.1 nmap-libdnet-1.12 ipv6
Compiled without:
Available nsock engines: epoll poll select
chundurur@chundurur-virtual-machine:~$
```

- Example: `nmap -v 192.168.1.1`
- Description: An advanced security analysis tool that scans networks and systems for

open ports, services, and other security vulnerabilities.

Command: `hostname`

```
chunduru@chunduru-virtual-machine:~$ hostname
chunduru-virtual-machine
chunduru@chunduru-virtual-machine:~$
```

- Example: `hostname`
- Description: Displays the system's host name.

Command: `hostnamectl`

```
chunduru@chunduru-virtual-machine:~$ hostnamectl
Static hostname: chunduru-virtual-machine
Icon name: computer-vm
Chassis: vm
Machine ID: 77cebc6a4a1f4c2d9a566908ea086a8a
Boot ID: 9b5e39221bf84e7095454b3d32461e7c
Virtualization: vmware
Operating System: Ubuntu 22.04.5 LTS
Kernel: Linux 6.8.0-136-generic
Architecture: x86-64
Hardware Vendor: VMware, Inc.
Hardware Model: VMware Virtual Platform
chunduru@chunduru-virtual-machine:~$
```

- Example: `hostnamectl set-hostname myhostname`
- Description: Used for host name management, including viewing and changing the system's hostname.

Working with Systemd commands

`systemctl start`

```
chunduru@chunduru-virtual-machine:~$ sudo systemctl start apache2
```

- Example: `systemctl start httpd`
- Description: Starts a specified unit file.

`systemctl status`

```
chunduru@chunduru-virtual-machine:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/lib/systemd/system/apache2.service; enabled; vendor preset: enabled)
   Active: active (running) since Sun 2026-08-09 18:15:21 IST; 3h 4min ago
     Docs: https://httpd.apache.org/docs/2.4/
    Main PID: 975 (apache2)
      Tasks: 55 (Limit: 2202)
    Memory: 2.7M
      CPU: 1.275s
    CGroup: /system.slice/apache2.service
            └─ 975 /usr/sbin/apache2 -k start
              977 /usr/sbin/apache2 -k start
              1008 /usr/sbin/apache2 -k start

Aug 09 18:15:20 chunduru-virtual-machine systemd[1]: Starting The Apache HTTP Server...
Aug 09 18:15:21 chunduru-virtual-machine apache2[915]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set the 'ServerName' directive globally to
Aug 09 18:15:21 chunduru-virtual-machine systemd[1]: Started The Apache HTTP Server.
lines 1-16/16 (END)
```

- Example: `systemctl status httpd`
- Description: Displays the current status of a unit file.

systemctl restart

```
chundurur@chundurur-virtual-machine:~$ systemctl restart apache2
chundurur@chundurur-virtual-machine:~$
```

- Example: `systemctl restart httpd`
- Description: Restarts a unit file, often used after making configuration changes.

systemctl stop

```
chundurur@chundurur-virtual-machine:~$ systemctl stop apache2
chundurur@chundurur-virtual-machine:~$
```

- Example: `systemctl stop httpd`
- Description: Stops a running unit file.

systemctl enable

```
chundurur@chundurur-virtual-machine:~$ systemctl enable apache2
Synchronizing state of apache2.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable apache2
chundurur@chundurur-virtual-machine:~$
```

- Example: `systemctl enable httpd`
- Description: Enables a unit file to start automatically on system boot.

systemctl disable

```
chundurur@chundurur-virtual-machine:~$ systemctl disable apache2
Synchronizing state of apache2.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install disable apache2
update-rc.d: error: Permsisston denied
chundurur@chundurur-virtual-machine:~$
```

- Example: `systemctl disable httpd`
- Description: Disables a unit file from starting automatically on boot.

systemctl list-units

```
chundurur@chundurur-virtual-machine:~$ systemctl list-units
```

UNIT	LOAD	ACTIVE	SUB	DESCRIPTION
proc-sys-fs-binfmt_misc.autonmount	loaded	active	running	Arbitrary Executable File Fornats File System Automount Point
sys-devices-pci0000:00-0000:00:10-0-host15-target15:0:0-15:0:0-0-block-sda-sda1.device	loaded	active	plugged	Vmware_Virtual_S_1
sys-devices-pci0000:00-0000:00:10-0-host15-target15:0:0-15:0:0-0-block-sda-sda2.device	loaded	active	plugged	Vmware_Virtual_S_EFI\x20System\x20Partition
sys-devices-pci0000:00-0000:00:10-0-host15-target15:0:0-15:0:0-0-block-sda-sda3.device	loaded	active	plugged	Vmware_Virtual_S_3
sys-devices-pci0000:00-0000:00:10-0-host15-target15:0:0-15:0:0-0-block-sda.device	loaded	active	plugged	Vmware_Virtual_S_5
sys-devices-pci0000:00-0000:00:11-0-0000:02:01-0-net-ens33.device	loaded	active	plugged	82545ER Gigabit Ethernet Controller (Copper) (PRO/1000 MT Single Port A
sys-devices-pci0000:00-0000:00:11-0-0000:02:02-0-sound-card0-controlC0.device	loaded	active	plugged	/sys/devices/pci0000:00/0000:00:11.0/0000:02:02.0/sound/card0/controlC0
sys-devices-pci0000:00-0000:00:11-0-0000:02:04-0-ata3-host2-target2:0:0-2:0:0-0-block-sr0.device	loaded	active	plugged	Vmware_Virtual_SATA_CDROM_Drive CDROM
sys-devices-pci0000:00-0000:00:11-0-0000:02:04-0-ata3-host3-target3:0:0-3:0:0-0-block-sr1.device	loaded	active	plugged	Vmware_Virtual_SATA_CDROM_Drive Ubuntu_22.04.5_LTS_and64
sys-devices-platform-floppy-0-block-fd0.device	loaded	active	plugged	/sys/devices/platform/floppy.0/block/fd0
sys-devices-platform-serial8250-serial8250:0-0-tty-ttyS0.device	loaded	active	plugged	/sys/devices/platform/serial8250/serial8250:0/serial8250:0.0/tty/ttyS0
sys-devices-platform-serial8250-serial8250:0-1-tty-ttyS1.device	loaded	active	plugged	/sys/devices/platform/serial8250/serial8250:0/serial8250:0.1/tty/ttyS1

- Example: `systemctl list-units`
- Description: Lists all currently loaded unit files.

systemctl set-default

```
chundurur@chundurur-virtual-machine:~$ sudo systemctl set-default graphical.target
Created symlink /etc/systemd/system/default.target → /lib/systemd/system/graphical.target.
chundurur@chundurur-virtual-machine:~$
```

- Example: `systemctl set-default graphical.target`
- Description: Sets the default target that the system boots into.

systemctl get-default

```
chundurur@chundurur-virtual-machine:~$ systemctl get-default
graphical.target
chundurur@chundurur-virtual-machine:~$
```

- Example: `systemctl get-default`
- Description: Displays the current default target.

systemctl cat

```
chundurur@chundurur-virtual-machine:~$ systemctl cat apache2
# /lib/systemd/system/apache2.service
[Unit]
Description=The Apache HTTP Server
After=network.target remote-fs.target nss-lookup.target
Documentation=https://httpd.apache.org/docs/2.4/

[Service]
Type=forking
Environment=APACHE_STARTED_BY_SYSTEMD=true
ExecStart=/usr/sbin/apachectl start
ExecStop=/usr/sbin/apachectl graceful-stop
ExecReload=/usr/sbin/apachectl graceful
KillMode=mixed
PrivateTmp=true
Restart=on-abort

[Install]
WantedBy=multi-user.target
chundurur@chundurur-virtual-machine:~$
```

- Example: `systemctl cat httpd`
- Description: Shows the configuration of a unit file.

systemctl show

```
chundurur@chundurur-virtual-machine:~$ systemctl show apache2
Type=forking
Restart=on-abort
NotifyAccess=none
RestartUsec=100ms
TimeoutStartUsec=1min 30s
TimeoutStopUsec=1min 30s
TimeoutAbortUsec=1min 30s
TimeoutStartFailureMode=terminate
TimeoutStopFailureMode=terminate
RuntimeMaxUsec=infinity
WatchdogUsec=0
WatchdogTimestamp=n/a
WatchdogTimestampMonotonic=0
RootDirectoryStartOnly=no
RemainAfterExit=no
GuessMainPID=yes
MainPID=0
ControlPID=0
FileDescriptorStoreMax=0
NFileDescriptorStore=0
StatusErrno=0
Result=success
ReloadResult=success
CleanResult=success
UID=[not set]
GID=[not set]
NRestarts=0
```

- Example: `systemctl show httpd`
- Description: Displays all the options set for a unit file.

systemctl edit

```
chundurur@chundurur-virtual-machine:~$ systemctl edit apache2
Failed to create directories for "/etc/systemd/system/apache2.service.d/override.conf": Permission denied
chundurur@chundurur-virtual-machine:~$ nano /etc/apache2/apache2.conf
chundurur@chundurur-virtual-machine:~$ ls -l /etc/apache2/apache2.conf
-rw-r--r-- 1 root root 7224 Jul  6 22:11 /etc/apache2/apache2.conf
```

- Example: `systemctl edit httpd`
- Description: Opens an editor to modify a unit file.

systemctl daemon-reload

```
chundurur@chundurur-virtual-machine:~$ systemctl daemon-reload
chundurur@chundurur-virtual-machine:~$
```

- Example: `systemctl daemon-reload`
- Description: Instructs the systemctl main process to reload its configuration.

systemctl isolate

```
chundurur@chundurur-virtual-machine:~$ systemctl isolate graphical.target
chundurur@chundurur-virtual-machine:~$
```

- Example: `systemctl isolate graphical.target`
- Description: Switches the system to a different target.

systemctl list-dependencies

```
chundurur@chundurur-virtual-machine:~$ systemctl list-dependencies graphical.target
graphical.target
├─accounts-daemon.service
├─apport.service
├─gdm.service
├─power-profiles-daemon.service
├─switcheroo-control.service
├─systemd-update-utmp-runlevel.service
├─udisks2.service
├─multi-user.target
├─anacron.service
├─apache2.service
├─apport.service
├─avahi-daemon.service
├─console-setup.service
├─cron.service
├─cups-browsed.service
├─cups.path
├─cups.service
├─dbus.service
├─dmesg.service
├─e2scrub_reap.service
├─grub-common.service
├─grub-initrd-fallback.service
├─irqbalance.service
├─kerneloops.service
├─ModemManager.service
├─mysql.service
├─networkd-dispatcher.service
├─NetworkManager.service
├─nginx.service
├─open-vm-tools.service
├─openvpn.service
```

- Example: `systemctl list-dependencies graphical.target`
- Description: Lists all dependencies of a specific target.

Managing Software Commands

Command: rpm

```
chundurur@chundurur-virtual-machine:~$ rpm --version
RPM version 4.17.0
chundurur@chundurur-virtual-machine:~$
```

- Description: Originally used for installing packages, `rpm` is now mainly used for querying information about installed packages.
- Example: `rpm -q package_name` - This command queries details about a specific package.

Command: yum

```
chundurur@chundurur-virtual-machine:~$ yum install package-name
Command 'yum' not found, did you mean:
  command 'gum' from snap gum (0.13.0)
  command 'uum' from deb freewnn-jserver (1.1.1~a021+cv520130302-7build1)
  command 'zum' from deb perforate (1.2-5.1)
  command 'yum4' from deb nextgen-yum4 (4.5.2-6)
  command 'sum' from deb coreutils (8.32-4.1ubuntu1.3)
  command 'num' from deb quickcal (2.4-1)
See 'snap info <snapname>' for additional versions.
```

- Description: `yum` is a software package manager used in Red Hat-based environments for installing, updating, and managing packages.
- Example: `yum install package_name` - This command installs a specific package.

Command: dnf

```
chundurur@chundurur-virtual-machine:~$ dnf --version
/usr/lib/python3/dist-packages/dnf/const.py:22: DeprecationWarning: The distutils package is deprecated and slated for removal in Python 3.12. Use setuptools or check PEP 632 for potential alternatives
  import distutils.sysconfig
/usr/lib/python3/dist-packages/dnf/const.py:22: DeprecationWarning: The distutils.sysconfig module is deprecated, use sysconfig instead
  import distutils.sysconfig
4.5.2
chundurur@chundurur-virtual-machine:~$
```

- Description: Similar to `yum`, `dnf` is a modern package manager for Red Hat-based systems, offering improved performance and package management capabilities.
- Example: `dnf update` - This command updates all installed packages to their latest versions.

Command: apt

```
chundurur@chundurur-virtual-machine:~$ sudo apt install dnf
```

- Description: `apt` is a command-line tool used in Ubuntu and other Debian-based systems for handling packages, including installation, updating, and removal.

- Example: `apt install package_name` - This command installs a specific package.

Command: `apt-cache`

```
chundurug@chundurug-virtual-machine:~$ apt-cache show apache2
Package: apache2
Priority: optional
Section: web
Installed-Size: 537
Maintainer: Ubuntu Developers <ubuntu-devel-discuss@lists.ubuntu.com>
Architecture: amd64
Version: 2.4.52-1ubuntu4.23
Recommends: ssl-cert
Replaces: apache2.2-bin, apache2.2-common
Suggests: apache2-doc, apache2-suexec-pristine | apache2-suexec-custom, www-browser, ufw
Provides: httpd, httpd-cgi
Depends: apache2-bin (= 2.4.52-1ubuntu4.23), apache2-data (= 2.4.52-1ubuntu4.23), apache2-utils (= 2.4.52-1ubuntu4.23), lsb-base, mime-support, perl:any, procps
Conflicts: apache2.2-bin, apache2.2-common
Pre-Depends: init-system-helpers (>= 1.54~)
Filename: pool/main/a/apache2/apache2_2.4.52-1ubuntu4.23_amd64.deb
Size: 97898
MD5sum: ecf7b30c2cb1180cd2b0330e340e9bf8
SHA1: 029c40b6e58e28a2ff95f41202d63be29fb6c72d
SHA256: 43780362eb783bfc77cf8df02c63bdd9b51b27467d120d626dc3bc305e3a8a20
SHA512: a966cfd159d62a03591868e5441bf3ccf840fa102d13241a9d164f15a466449ba6b754dc9518d87ea8e0dc002003bb8e46a7d27899bed425d735738bc4ca0b1
Homepage: https://httpd.apache.org/
Description-en: Apache HTTP Server
The Apache HTTP Server Project's goal is to build a secure, efficient and
extensible HTTP server as standards-compliant open source software. The
result has long been the number one web server on the Internet.
.
Installing this package results in a full installation, including the
configuration files, init scripts and support scripts.
Description-md5: d02426bc360345e5acd45367716dc35c
```

- Description: `apt-cache` is used for querying package information in Debian-based systems. It's especially useful for searching for packages containing specific files.
- Example: `apt-cache search keyword` - This command searches for packages related to a specific keyword.

Managing SSH Commands

Change SSH Port on Ubuntu:

```
chundurug@chundurug-virtual-machine:~$ sudo vim /etc/ssh/sshd-config
chundurug@chundurug-virtual-machine:~$
```

- Command: `sudo vim /etc/ssh/sshd_config`
- Description: Open the SSH configuration file using `vim` to change the SSH port.

Update SSH Port to 2022:

```
chundurug@chundurug-virtual-machine:~$ sudo snap install port
port 1.10.0 from Hunter Miller (arhyn) installed
chundurug@chundurug-virtual-machine:~$ port 22
/snap/port/78/port/port: /lib/x86_64-linux-gnu/libc.so.6: version 'GLIBC_2.30' not found (required by /snap/port/78/usr/lib/x86_64-linux-gnu/libnspr4.so)
/snap/port/78/port/port: /lib/x86_64-linux-gnu/libm.so.6: version 'GLIBC_2.29' not found (required by /snap/port/78/usr/lib/x86_64-linux-gnu/libasound.so.2)
```

- Command: Change the line `Port 22` to `Port 2022` in the file.
- Description: In the SSH configuration file, update the default port from 22 to 2022.

Restart SSH Service:

```

chunduru@chunduru-virtual-machine:~$ sudo systemctl restart ssh
chunduru@chunduru-virtual-machine:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: enabled)
   Active: active (running) since Sun 2026-08-09 22:42:24 IST; 13s ago
     Docs: man:sshd(8)
           man:sshd_config(5)
   Process: 14424 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
  Main PID: 14425 (sshd)
    Tasks: 1 (limit: 2202)
   Memory: 1.7M
      CPU: 40ms
   CGroup: /system.slice/ssh.service
           └─14425 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Aug 09 22:42:24 chunduru-virtual-machine systemd[1]: Starting OpenBSD Secure Shell server...
Aug 09 22:42:24 chunduru-virtual-machine sshd[14425]: Server listening on 0.0.0.0 port 22.
Aug 09 22:42:24 chunduru-virtual-machine sshd[14425]: Server listening on :: port 22.
Aug 09 22:42:24 chunduru-virtual-machine systemd[1]: Started OpenBSD Secure Shell server.
chunduru@chunduru-virtual-machine:~$

```

- **Command:** `sudo systemctl restart sshd`
- **Description:** Restart the SSH daemon to apply the new configuration.

Allow Port 2022 in Firewall:

```

chunduru@chunduru-virtual-machine:~$ sudo ufw allow 2022/tcp
Rules updated
Rules updated (v6)
chunduru@chunduru-virtual-machine:~$

```

- **Command:** `sudo ufw allow 2022/tcp`
- **Description:** Update the Ubuntu firewall to allow TCP connections on port 2022.

Check SSH Service Status:

```

chunduru@chunduru-virtual-machine:~$ sudo systemctl status sshd
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: enabled)
   Active: active (running) since Sun 2026-08-09 22:42:24 IST; 2min 50s ago
     Docs: man:sshd(8)
           man:sshd_config(5)
   Process: 14424 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
  Main PID: 14425 (sshd)
    Tasks: 1 (limit: 2202)
   Memory: 1.7M
      CPU: 40ms
   CGroup: /system.slice/ssh.service
           └─14425 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Aug 09 22:42:24 chunduru-virtual-machine systemd[1]: Starting OpenBSD Secure Shell server...
Aug 09 22:42:24 chunduru-virtual-machine sshd[14425]: Server listening on 0.0.0.0 port 22.
Aug 09 22:42:24 chunduru-virtual-machine sshd[14425]: Server listening on :: port 22.
Aug 09 22:42:24 chunduru-virtual-machine systemd[1]: Started OpenBSD Secure Shell server.
chunduru@chunduru-virtual-machine:~$

```

- **Command:** `sudo systemctl status sshd`
- **Description:** Verify that the SSH service is active and listening on port 2022.

Create SSH Session from Windows:

```

chunduru@chunduru-virtual-machine:~$ hostname -I
192.168.85.128
chunduru@chunduru-virtual-machine:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: enabled)
   Active: active (running) since Sun 2026-08-09 22:42:24 IST; 4min 54s ago
     Docs: man:sshd(8)
           man:sshd_config(5)
   Process: 14424 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
    Main PID: 14425 (sshd)
       Tasks: 1 (limit: 2202)
      Memory: 1.7M
         CPU: 40ms
    CGroup: /system.slice/ssh.service
            └─14425 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Aug 09 22:42:24 chunduru-virtual-machine systemd[1]: Starting OpenBSD Secure Shell server...
Aug 09 22:42:24 chunduru-virtual-machine sshd[14425]: Server listening on 0.0.0.0 port 22.
Aug 09 22:42:24 chunduru-virtual-machine sshd[14425]: Server listening on :: port 22.
Aug 09 22:42:24 chunduru-virtual-machine systemd[1]: Started OpenBSD Secure Shell server.
chunduru@chunduru-virtual-machine:~$ sudo ss -tlnp | grep 2022
chunduru@chunduru-virtual-machine:~$

```

- Description: Use MobaXterm on Windows to create a new SSH session.
- Steps:
 - Open MobaXterm and select to create a new SSH session.
 - Enter the remote host IP address (e.g., 192.168.29.142).
 - Specify the username (student in this case).
 - Set the port to 2022.
 - Click OK to initiate the connection, then enter the password when prompted.

Managing Time commands

Command: date

```

chunduru@chunduru-virtual-machine:~$ date
Sunday 09 August 2026 09:56:34 PM IST
chunduru@chunduru-virtual-machine:~$

```

- Example: date
- Description: Displays the current date and time of the system. It's a basic and long-standing command in many operating systems.

Command: hwclock

```

chunduru@chunduru-virtual-machine:~$ sudo hwclock --show
2026-08-09 21:58:15.983846+05:30
chunduru@chunduru-virtual-machine:~$

```

- Example: hwclock --show
- Description: Used for managing and viewing the hardware clock. It can also be used to

synchronize the hardware clock with the system clock.

Command: `timedatectl`

```
chundurur@chundurur-virtual-machine:~$ timedatectl
      Local time: Sun 2026-08-09 21:59:08 IST
      Universal time: Sun 2026-08-09 16:29:08 UTC
          RTC time: Sun 2026-08-09 16:29:08
      Time zone: Asia/Kolkata (IST, +0530)
System clock synchronized: yes
      NTP service: active
      RTC in local TZ: no
```

- Example: `timedatectl status`
- Description: A modern utility for managing various aspects of time synchronization and settings. It can display and set time, date, and timezone, among other features.

Command: `chronyc`

```
chundurur@chundurur-virtual-machine:~$ chronyc tracking
Reference ID    : ACECB40F (172-236-180-15.ip.linodeusercontent.com)
Stratum        : 6
Ref time (UTC) : Sun Aug 09 16:32:50 2026
System time    : 0.000002158 seconds fast of NTP time
Last offset    : -0.004206022 seconds
RMS offset     : 0.004206022 seconds
Frequency      : 16.162 ppm slow
Residual freq  : -38.688 ppm
Skew           : 1000000.000 ppm
Root delay     : 0.035928104 seconds
Root dispersion : 3.836517572 seconds
Update interval : 1.5 seconds
Leap status    : Normal
chundurur@chundurur-virtual-machine:~$
```

- Example: `chronyc tracking`
- Description: Interfaces with the `chronyd` time-synchronizing service, providing detailed information about the current synchronization status and more control over the `chronyd` service.

Process Management Commands

Command: `top`

```
chundurug@chunduru-virtual-machine:~$ top
top - 22:03:34 up 3:48, 1 user, load average: 0.21, 0.19, 0.13
Tasks: 300 total, 1 running, 299 sleeping, 0 stopped, 0 zombie
%cpu(s): 1.7 us, 0.8 sy, 0.0 ni, 97.1 id, 0.0 wa, 0.0 hi, 0.3 si, 0.0 st
Mem Mem: 1920.2 total, 259.3 free, 828.7 used, 832.1 buff/cache
Mem Swap: 2140.0 total, 1527.4 free, 612.6 used, 890.3 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR     S    %CPU  %MEM     time+ COMMAND
 1188 mysql      20   0 1784516 10376  9420 S    1.7   0.5   3:09.62 mysqld
 1825 chunduru  20   0 4292536 186600 72384 S    1.7   9.5   3:58.25 gnome-shell
 598 root        20   0 317384   7000   6252 S    0.7   0.4   1:08.17 vntoolsd
 938 tomcat     20   0 3060568 79772 21572 S    0.7   4.1   0:59.76 Java
2131 chunduru  20   0 149390 36728 19936 S    0.7   1.9   1:05.15 vntoolsd
8597 chunduru  20   0 13240 4336 3376 R    0.7   0.2   0:00.07 top
 814 root        20   0 16504 4084 4008 S    0.3   0.2   0:00.17 wpa_supplicant
2228 chunduru  20   0 163620 7220 6292 S    0.3   0.4   0:16.52 dbus-engine-sin
2495 chunduru  20   0 560548 45768 35064 S    0.3   2.3   1:48.75 gnome-terminal-
5611 root        20   0 0 0 0 I    0.3   0.0   0:01.39 kworker/u258:2-events_freezable_power_
7075 root        20   0 0 0 0 I    0.3   0.0   0:00.20 kworker/0:1-npt_poll_0
 1 root        20   0 168116 12504 7988 S    0.0   0.6   0:10.44 systemd
 2 root        20   0 0 0 0 S    0.0   0.0   0:00.10 kthreadd
 3 root        20   0 0 0 0 S    0.0   0.0   0:00.00 pool_workqueue_release
 4 root        0 -20 0 0 0 I    0.0   0.0   0:00.09 kworker/R-rcu_g
 5 root        0 -20 0 0 0 I    0.0   0.0   0:00.00 kworker/R-rcu_p
 6 root        0 -20 0 0 0 I    0.0   0.0   0:00.00 kworker/R-slub_
 7 root        0 -20 0 0 0 I    0.0   0.0   0:00.00 kworker/R-netns
11 root        0 -20 0 0 0 I    0.0   0.0   0:00.13 kworker/u256:0-ext4-rsv-conversion
12 root        0 -20 0 0 0 I    0.0   0.0   0:00.00 kworker/R-mm_pe
13 root        0 -20 0 0 0 I    0.0   0.0   0:00.00 rcu_tasks_kthread
14 root        0 -20 0 0 0 I    0.0   0.0   0:00.00 rcu_tasks_rude_kthread
15 root        0 -20 0 0 0 I    0.0   0.0   0:00.00 rcu_tasks_trace_kthread
16 root        0 -20 0 0 0 S    0.0   0.0   0:00.45 ksoftirqd/0
17 root        0 -20 0 0 0 I    0.0   0.0   0:02.24 rcu_preempt
```

- Example: `top`
- Description: Monitors all processes and the general health of the system from a performance perspective.

Command: `ps`

```
chundurug@chunduru-virtual-machine:~$ ps aux
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1  0.0  0.6 168116 12504 ?        Ss   18:15   0:10 /sbin/init auto noprompt splash
root         2  0.0  0.0 0          0 ?        S    18:15   0:00 [kthreadd]
root         3  0.0  0.0 0          0 ?        S    18:15   0:00 [pool_workqueue_release]
root         4  0.0  0.0 0          0 ?        I<   18:15   0:00 [kworker/R-rcu_g]
root         5  0.0  0.0 0          0 ?        I<   18:15   0:00 [kworker/R-rcu_p]
root         6  0.0  0.0 0          0 ?        I<   18:15   0:00 [kworker/R-slub_]
root         7  0.0  0.0 0          0 ?        I<   18:15   0:00 [kworker/R-netns]
root        11  0.0  0.0 0          0 ?        I    18:15   0:00 [kworker/u256:0-ext4-rsv-conversion]
root        12  0.0  0.0 0          0 ?        I<   18:15   0:00 [kworker/R-mm_pe]
root        13  0.0  0.0 0          0 ?        I    18:15   0:00 [rcu_tasks_kthread]
root        14  0.0  0.0 0          0 ?        I    18:15   0:00 [rcu_tasks_rude_kthread]
root        15  0.0  0.0 0          0 ?        I    18:15   0:00 [rcu_tasks_trace_kthread]
root        16  0.0  0.0 0          0 ?        S    18:15   0:00 [ksoftirqd/0]
root        17  0.0  0.0 0          0 ?        I    18:15   0:02 [rcu_preempt]
root        18  0.0  0.0 0          0 ?        S    18:15   0:00 [migration/0]
root        19  0.0  0.0 0          0 ?        S    18:15   0:00 [idle_inject/0]
root        20  0.0  0.0 0          0 ?        S    18:15   0:00 [cpuhp/0]
root        21  0.0  0.0 0          0 ?        S    18:15   0:00 [cpuhp/1]
root        22  0.0  0.0 0          0 ?        S    18:15   0:00 [idle_inject/1]
root        23  0.0  0.0 0          0 ?        S    18:15   0:00 [migration/1]
root        24  0.0  0.0 0          0 ?        S    18:15   0:01 [ksoftirqd/1]
root        29  0.0  0.0 0          0 ?        S    18:15   0:00 [kdevtmpfs]
root        30  0.0  0.0 0          0 ?        I<   18:15   0:00 [kworker/R-inet_]
root        31  0.0  0.0 0          0 ?        S    18:15   0:00 [kauditd]
root        33  0.0  0.0 0          0 ?        S    18:15   0:00 [khungtaskd]
root        34  0.0  0.0 0          0 ?        S    18:15   0:00 [oom_reaper]
root        37  0.0  0.0 0          0 ?        I<   18:15   0:00 [kworker/R-write]
root        38  0.0  0.0 0          0 ?        S    18:15   0:06 [kcompactd0]
root        39  0.0  0.0 0          0 ?        SN   18:15   0:00 [ksmd]
root        42  0.0  0.0 0          0 ?        SN   18:15   0:00 [khugepaged]
root        43  0.0  0.0 0          0 ?        I<   18:15   0:00 [kworker/R-kinte]
```

- Example: `ps aux`

- Description: Shows all processes and their properties; useful for monitoring system processes.

Command: `jobs`

```
chunduru@chunduru-virtual-machine:~$ jobs
chunduru@chunduru-virtual-machine:~$
```

- Example: `jobs`
- Description: Lists the jobs (processes) started by the user in the shell, allowing for management of interactive processes.

Command: `fg`

```
chunduru@chunduru-virtual-machine:~$ fg %1
bash: fg: %1: no such job
chunduru@chunduru-virtual-machine:~$
```

- Example: `fg %1`
- Description: Brings a job to the foreground, where %1 is the job number.

Command: `bg`

```
chunduru@chunduru-virtual-machine:~$ jobs
chunduru@chunduru-virtual-machine:~$ bg %1
bash: bg: %1: no such job
chunduru@chunduru-virtual-machine:~$
```

- Example: `bg %1`
- Description: Sends a job to run in the background.

Command: `nice`

```
chunduru@chunduru-virtual-machine:~$ nice -n 10
nice: a command must be given with an adjustment
Try 'nice --help' for more information.
chunduru@chunduru-virtual-machine:~$
```

- Example: `nice -n 10 command`
- Description: Starts a process with a defined niceness (priority), where `command` is the command to be run with a priority.

Command: renice

```
chundurur@chundurur-virtual-machine:~$ ps
  PID TTY          TIME CMD
 2513 pts/0    00:00:01 bash
 8613 pts/0    00:00:00 ps
chundurur@chundurur-virtual-machine:~$ sudo renice 10 -p 1234
renice: failed to get priority for 1234 (process ID): No such process
chundurur@chundurur-virtual-machine:~$ ps -0 pid,ni,comm -p 1234
error: process ID out of range

Usage:
ps [options]

Try 'ps --help <simple|list|output|threads|misc|all>'
or 'ps --help <s|l|o|t|m|a>'
for additional help text.

For more details see ps(1).
chundurur@chundurur-virtual-machine:~$
```

- Example: `renice -n 10 -p 1234`
- Description: Changes the priority of an already running process, where 1234 is the Process ID (PID).

Command: kill

```
chundurur@chundurur-virtual-machine:~$ kill 1234
bash: kill: (1234) - No such process
chundurur@chundurur-virtual-machine:~$
```

- Example: `kill 1234`
- Description: Sends a signal to a process (usually to terminate it), where 1234 is the Process ID (PID).

Command: killall

```
chundurur@chundurur-virtual-machine:~$ killall
Usage: killall [OPTION]... [--] NAME...
    killall -l, --list
    killall -V, --version

-e,--exact          require exact match for very long names
-I,--ignore-case    case insensitive process name match
-g,--process-group  kill process group instead of process
-y,--younger-than   kill processes younger than TIME
-o,--older-than     kill processes older than TIME
-i,--interactive    ask for confirmation before killing
-l,--list           list all known signal names
-q,--quiet          don't print complaints
-r,--regex          interpret NAME as an extended regular expression
-s,--signal SIGNAL  send this signal instead of SIGTERM
-u,--user USER      kill only process(es) running as USER
-v,--verbose        report if the signal was successfully sent
-V,--version        display version information
-w,--wait           wait for processes to die
-n,--ns PID         match processes that belong to the same namespaces
                    as PID
-Z,--context REGEXP kill only process(es) having context
                    (must precede other arguments)
```

- Example: `killall processname`

- Description: Terminates all processes with the given name.

Scheduling Tasks Commands

at Command

```
chunduru@chunduru-virtual-machine:~$ at
Garbled time
chunduru@chunduru-virtual-machine:~$
```

- Description: Allows you to schedule a job to run at a specific time.
- Example: `at 10:30` schedules a job to run at 10:30.

atq Command

```
chunduru@chunduru-virtual-machine:~$ atq
2      Mon Aug 10 22:00:00 2026 a chunduru
chunduru@chunduru-virtual-machine:~$
```

- Description: Displays a list of all jobs currently scheduled.
- Example: `atq` shows all scheduled jobs with their job numbers.

atrm Command

```
chunduru@chunduru-virtual-machine:~$ atrm
Usage: at [-V] [-q x] [-f file] [-u username] [-mMlbv] timespec ...
       at [-V] [-q x] [-f file] [-u username] [-mMlbv] -t time
       at -c job ...
       at [-V] -l [-o timeformat] [job ...]
       atq [-V] [-q x] [-o timeformat] [job ...]
       at [-rd] job ...
       atrm [-V] job ...
       batch
chunduru@chunduru-virtual-machine:~$
```

- Description: Used to remove scheduled jobs.
- Example: `atrm 5` removes the job with the job number 5.

Editing Cron Configuration with `crontab -e`

```
chunduru@chunduru-virtual-machine:~$ crontab -e
no crontab for chunduru - using an empty one

Select an editor. To change later, run 'select-editor'.
 1. /bin/nano          <---- easiest
 2. /usr/bin/vim.basic
 3. /usr/bin/vim.tiny
 4. /bin/ed

Choose 1-4 [1]:
```

- Description: Opens the cron configuration for editing, allowing you to schedule jobs to run periodically.

- Example: `crontab -e` opens the cron configuration file in a text editor.

Working with Systemctl Timers Using `systemctl`

```
chunduru@chunduru-virtual-machine:~$ systemctl list-timers

```

NEXT	LEFT	LAST	PASSED	UNIT	ACTIVATES
Sun 2026-08-09 22:34:12 IST	14min left	Sun 2026-08-09 21:33:12 IST	46min ago	anacron.timer	anacron.service
Sun 2026-08-09 23:41:36 IST	1h 21min left	Sun 2026-08-09 22:02:40 IST	17min ago	fwupd-refresh.timer	fwupd-refresh.service
Mon 2026-08-10 00:00:00 IST	1h 39min left	n/a	n/a	dpkg-db-backup.timer	dpkg-db-backup.service
Mon 2026-08-10 00:00:00 IST	1h 39min left	Sun 2026-08-09 18:15:19 IST	4h 4min ago	logrotate.timer	logrotate.service
Mon 2026-08-10 01:00:56 IST	2h 40min left	Tue 2026-08-04 11:15:45 IST	5 days ago	fstrim.timer	fstrim.service
Mon 2026-08-10 01:47:23 IST	3h 27min left	Sun 2026-08-09 21:25:52 IST	54min ago	man-db.timer	man-db.service
Mon 2026-08-10 05:37:40 IST	7h left	Sun 2026-08-09 18:37:06 IST	3h 42min ago	plocate-updatedb.timer	plocate-updatedb.service
Mon 2026-08-10 06:35:50 IST	8h left	Sun 2026-08-09 18:47:38 IST	3h 32min ago	apt-daily-upgrade.timer	apt-daily-upgrade.service
Mon 2026-08-10 06:56:15 IST	8h left	Sun 2026-08-09 18:23:12 IST	3h 56min ago	motd-news.timer	motd-news.service
Mon 2026-08-10 13:50:56 IST	15h left	Sun 2026-08-09 21:35:47 IST	44min ago	apt-daily.timer	apt-daily.service
Mon 2026-08-10 18:20:19 IST	20h left	Sun 2026-08-09 18:20:19 IST	3h 59min ago	update-notifier-download.timer	update-notifier-download.service
Mon 2026-08-10 18:30:12 IST	20h left	Sun 2026-08-09 18:30:12 IST	3h 49min ago	systemd-tmpfiles-clean.timer	systemd-tmpfiles-clean.service
Fri 2026-08-14 14:32:55 IST	4 days left	Tue 2026-08-04 11:15:48 IST	5 days ago	update-notifier-motd.timer	update-notifier-motd.service
Sun 2026-08-16 03:10:35 IST	6 days left	Sun 2026-08-09 18:15:51 IST	4h 4min ago	e2scrub_all.timer	e2scrub_all.service

```

14 timers listed.
Pass --all to see loaded but inactive timers, too.

```

- Description: Used to manage and interact with `systemctl` timers, a more systemwide approach for scheduling tasks.
- Example: `systemctl list-timers` shows all active timers.

Reading Log Files Commands

Command: `journalctl`

```
chunduru@chunduru-virtual-machine:~$ journalctl -u nginx-service
-- No entries --
chunduru@chunduru-virtual-machine:~$
```

- Description: This command is used to check the contents of the `systemd` journal, which is a centralized logging system for Linux.
- Example: `journalctl -u nginx.service`
 - This example shows how to view logs for the Nginx service using `journalctl`.

Command: `tail`

```

chunduru@chunduru-virtual-machine:~$ tail -n 100 /var/log/syslog
Aug 9 21:37:16 chunduru-virtual-machine systemd[1]: dnsmasq.service: Deactivated successfully.
Aug 9 21:41:10 chunduru-virtual-machine dbus-daemon[742]: [system] Unknown group "power" in message bus configuration file
Aug 9 21:41:10 chunduru-virtual-machine dbus-daemon[742]: [system] Reloaded configuration
Aug 9 21:41:10 chunduru-virtual-machine dbus-daemon[742]: [system] Unknown group "power" in message bus configuration file
Aug 9 21:41:10 chunduru-virtual-machine dbus-daemon[742]: [system] Reloaded configuration
Aug 9 21:41:10 chunduru-virtual-machine dbus-daemon[742]: [system] Unknown group "power" in message bus configuration file
Aug 9 21:41:10 chunduru-virtual-machine dbus-daemon[742]: [system] Reloaded configuration
Aug 9 21:41:10 chunduru-virtual-machine dbus-daemon[742]: [system] Unknown group "power" in message bus configuration file
Aug 9 21:41:10 chunduru-virtual-machine dbus-daemon[742]: [system] Reloaded configuration
Aug 9 21:41:11 chunduru-virtual-machine dbus-daemon[742]: [system] Activating via systemd: service name='org.freedesktop.PackageKit' unit='packagekit.service' requested by ':1.179' (uid=0 pid=6677 comm="/usr/bin/gdbus call --system --dest org.freedesktop.PackageKit /org/freedesktop/PackageKit1 org.freedesktop.PackageKit1.Daemon...")
Aug 9 21:41:11 chunduru-virtual-machine PackageKit: daemon start
Aug 9 21:41:11 chunduru-virtual-machine dbus-daemon[742]: [system] Successfully activated service 'org.freedesktop.PackageKit'
Aug 9 21:41:11 chunduru-virtual-machine systemd[1]: Started PackageKit Daemon.
Aug 9 21:45:22 chunduru-virtual-machine NetworkManager[745]: <Info> [1786292122.5661] dhcp4 (ens33): state changed new lease, address=192.168.85.128
Aug 9 21:52:41 chunduru-virtual-machine systemd[1]: Starting Update APT News...
Aug 9 21:52:41 chunduru-virtual-machine systemd[1]: Starting Update the local ESM caches...
Aug 9 21:52:41 chunduru-virtual-machine systemd[1]: apt-news.service: Deactivated successfully.
Aug 9 21:52:41 chunduru-virtual-machine systemd[1]: Finished Update APT News.
Aug 9 21:52:42 chunduru-virtual-machine systemd[1]: esm-cache.service: Deactivated successfully.
Aug 9 21:52:42 chunduru-virtual-machine systemd[1]: Finished Update the local ESM caches.
Aug 9 21:59:08 chunduru-virtual-machine dbus-daemon[742]: [system] Activating via systemd: service name='org.freedesktop.timedate1' unit='dbus-org.freedesktop.timedate1.service' requested by ':1.184' (uid=1000 pid=7628 comm='timedatectl --label=unconfined')
Aug 9 21:59:08 chunduru-virtual-machine systemd[1]: Starting Time & Date Service...
Aug 9 21:59:08 chunduru-virtual-machine dbus-daemon[742]: [system] Successfully activated service 'org.freedesktop.timedate1'
Aug 9 21:59:08 chunduru-virtual-machine systemd[1]: Started Time & Date Service.
Aug 9 21:59:38 chunduru-virtual-machine systemd[1]: systemd-timedated.service: Deactivated successfully.
Aug 9 22:00:22 chunduru-virtual-machine NetworkManager[745]: <Info> [1786293022.5859] dhcp4 (ens33): state changed new lease, address=192.168.85.128
Aug 9 22:01:33 chunduru-virtual-machine systemd[1]: Starting Update APT News...
Aug 9 22:01:33 chunduru-virtual-machine systemd[1]: Starting Update the local ESM caches...
Aug 9 22:01:33 chunduru-virtual-machine systemd[1]: apt-news.service: Deactivated successfully.
Aug 9 22:01:34 chunduru-virtual-machine systemd[1]: Finished Update APT News.
Aug 9 22:01:34 chunduru-virtual-machine systemd[1]: esm-cache.service: Deactivated successfully.
Aug 9 22:01:34 chunduru-virtual-machine systemd[1]: Finished Update the local ESM caches.
Aug 9 22:02:39 chunduru-virtual-machine systemd[1]: Stopping Network Time Synchronization...
Aug 9 22:02:39 chunduru-virtual-machine systemd[1]: systemd-timesyncd.service: Deactivated successfully.
Aug 9 22:02:39 chunduru-virtual-machine systemd[1]: Stopped Network Time Synchronization.
Aug 9 22:02:39 chunduru-virtual-machine dbus-daemon[742]: [system] Unknown group "power" in message bus configuration file

```

- Description: The `tail` command is used to view the contents of log files, typically the last few lines.
- Example: `tail -n 100 /var/log/syslog`
 - This example displays the last 100 lines of the syslog file.

Command: less

```

chunduru@chunduru-virtual-machine:~$ less /var/log/apache2/access.log
/var/log/apache2/access.log: No such file or directory
chunduru@chunduru-virtual-machine:~$

```

- Description: Similar to `tail`, `less` is used for viewing log files but allows for more controlled navigation through the file.
- Example: `less /var/log/apache2/access.log`
 - This opens the Apache access log file in a scrollable interface for in-depth examination.

Command: logger

```

chunduru@chunduru-virtual-machine:~$ logger "System Backup Completed"
chunduru@chunduru-virtual-machine:~$ grep "System Backup Completed" /var/log/syslog
chunduru@chunduru-virtual-machine:~$

```

- Description: The `logger` command is a convenient tool for writing messages to the system log.
- Example: `logger "System backup completed"`
 - This example sends a custom message, "System backup completed," to the system log.